

Data management in substance use disorder treatment research: Implications from data harmonization of National Institute on Drug Abuse-funded randomized controlled trials

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Abstract

Background: Secondary analysis of data from completed randomized controlled trials is a critical and efficient way to maximize the potential benefits from past research. De-identified primary data from completed randomized controlled trials have been increasingly available in recent years; however, the lack of standardized data products is a major barrier to further use of these valuable data. Pre-statistical harmonization of data structure, variables, and codebooks across randomized controlled trials would facilitate secondary data analysis, including meta-analyses and comparative effectiveness studies. We describe a pre-statistical data harmonization initiative to standardize de-identified primary data from substance use disorder treatment randomized controlled trials funded by the National Institute on Drug Abuse available on the National Institute on Drug Abuse Data Share website.

Methods: Standardized datasets and codebooks with consistent data structures, variable names, labels, and definitions were developed for 36 completed randomized controlled trials. Common data domains were identified to bundle data files from individual randomized controlled trials according to relevant concepts. Variables were harmonized if at least two randomized controlled trials used the same instruments. The structures of the harmonized data were determined based on the feedback from clinical trialists and substance use disorder research experts.

Results: We have created a harmonized database of variables across 36 randomized controlled trials with a build-in label and a brief definition for each variable. Data files from the randomized controlled trials have been consistently categorized into eight domains (enrollment, demographics, adherence, adverse events, physical health measures, mental-behavioral-cognitive health measures, self-reported substance use measures, and biologic substance use measures). Standardized codebooks and concordance tables have also been developed to help identify instruments and variables of interest more easily.

Conclusion: The harmonized data of randomized controlled trials of substance use disorder treatments can potentially promote future secondary data analysis of completed randomized controlled trials, allowing combining data from multiple randomized controlled trials and provide guidance for future randomized controlled trials in substance use disorder treatment research.

Keywords

Randomized controlled trials, secondary data analysis, substance abuse treatment, data harmonization

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Background

Sharing de-identified primary data from randomized controlled trials (RCTs) has been increasingly promoted by the National Institutes of Health¹ and the wider scientific community.² Secondary analysis of data from the completed RCTs is a critical and efficient way to maximize the potential benefits from past research. RCTs for new treatments should reflect the lessons and implications from the past RCTs, and results from new trials should be comparable with existing studies. Greater data availability would allow investigators not involved in the primary research to confirm the findings of the original investigators, minimize the risk of selective non-reporting of results, reanalyze data using emerging statistical methods, explore relationships of theoretic and clinical interest that were not examined by primary investigators, and conduct more powerful tests of hypotheses through combining multiple datasets.³ Greater availability of standardized data from multiple RCTs would also advance the literature of individual patient data meta-analysis, which, compared with conventional aggregated data meta-analysis, has advantages, including increased precision of treatment effect estimates, analysis of heterogeneity using individual-level characteristics, and consistent analytic methods to reduce methodologic heterogeneity across studies.⁴⁻⁷ Harmonized and pooled data are cost-effective resources for comparative effectiveness studies by allowing comparisons across multiple active treatment arms.⁶ Furthermore, the harmonized data would allow analyses of treatment response in underrepresented population subgroups and exploration of treatment response heterogeneity in patients with different risk profiles. Such data can promote new directions for research by identifying population groups who would most benefit from interventions and hence contribute to ultimately improving clinical practice.

De-identified primary data from completed RCTs have been increasingly available in recent years in various biomedical research fields. There are several organizations making critical efforts to provide researchers with user-friendly and/or standardized RCT data for further scientific research. Some of the data repositories include the Inter-university Consortium of Political and Social Research, the Yale University of Open Data Access, the National Institute of Mental Health (NIMH) data archive,⁸ ClinicalStudyDataRequest.com,⁹ and Vivli Center for Global Clinical Research Data.¹⁰ In the field of substance use disorder treatment, the National Institute for Drug Abuse (NIDA) has made data from completed RCTs publicly available through the NIDA Data Share website¹¹ to encourage secondary analysis. The website includes RCTs

conducted under (1) the National Drug Abuse Treatment Clinical Trials Network, in which patients were recruited from a broad range of community-based substance use disorder treatment settings, and (2) the Division of Therapeutics and Medical Consequences program, in which patients were recruited through advertisements in various academic institutions, federal therapeutic development institutions, and the pharmaceutical and biotechnology sectors to test effectiveness of innovative substance use disorder pharmacotherapies. Because the United States has a high prevalence of substance use disorders, which are among the most important public health issues,^{12,13} there is a critical need for more efficacious treatments, and data from NIDA-funded RCTs are an extremely valuable resource for advancing the standard of evidence-based care.

In spite of the growing number of RCTs in the NIDA Data Share website, only a limited number of secondary analyses of these RCT data have been conducted and published by independent investigators who were not part of the primary RCT investigative teams.¹⁴ According to the list of the publications held by the NIDA Clinical Trials Network Dissemination Library, at the time of this writing, there were a total of 636 published reports using the NIDA Clinical Trials Network RCT data, including peer-reviewed journals, posters, and newsletters, most of which were secondary analyses of data based on NIDA Clinical Trials Network RCTs.¹⁵ However, only 34 (5.3%) of these documents were published by independent investigators. The difficulty of understanding and using RCT data from the NIDA Data Share website might be an important obstacle to independent analyses. Limited re-use of data that are already downloadable from a public website highlights the potential importance of creating user-friendly standardized data products with consistent variable names, labels, and codebooks.

The primary purpose of this article is to describe a data harmonization initiative to standardize de-identified primary data from NIDA-funded substance use disorder treatment RCTs available on the NIDA Data Share website. This initiative focuses on “pre-statistical” data harmonization which maintains the integrity of the original data elements while making them consistent across RCTs with diverse data collection tools, formats, and procedures.^{16,17} The pre-statistical data harmonization enables users to further synthesize information across RCTs through statistical data harmonization which relies on a specific analytical decision model. We aimed to develop standardized datasets of NIDA-funded RCTs that were deposited on the NIDA Data share website by September 2018.

Method

NIDA-funded RCTs

There were 36 primary datasets publicly available from the NIDA Data Share website by September 2018, including one RCT follow-up study (CTN0030A3). The scope of this pre-statistical data harmonization initiative was limited to those RCTs that compared two or more interventions or regimens for treatment of substance use disorder and that assessed substance use disorder outcomes. We excluded open-label trials, feasibility trials, safety/tolerability trials, non-randomized trials, and RCTs that were designed to assess non-substance use disorder outcomes, such as sexually transmitted disease infection from our data harmonization initiative.

Table 1 lists the 36 RCTs included in this data harmonization initiative. There were 23 RCTs conducted under the Clinical Trials Network and 13 RCTs conducted under the Division of Therapeutics and Medical Consequences program. By target substances, 16 RCTs assessed treatment of stimulant use disorders, including cocaine and methamphetamine, 9 RCTs assessed treatment of opioid use disorders, 8 RCTs assessed treatment of any substance use disorders, and 3 RCTs assessed treatment of nicotine use disorder. By intervention types, 23 RCTs tested the effectiveness of pharmacological interventions, while 13 RCTs tested the effectiveness of behavioral interventions. The sample sizes ranged from 62 (CTN0052) to 1285 (CTN0047).

Development of standardized data format

A number of instruments and other data elements were utilized in each completed RCT. Instruments included both standardized data elements in substance use disorder research fields, such as the Addiction Severity Index and the Clinical Opiate Withdrawal Scale and non-standardized data elements, such as demographics and urine toxicology results. The datasets in the standardized database were designed so that each data file contains only one type of instrument and that the number of data files corresponds to the number of instruments used in each RCT. Data files have been named in a consistent manner representing each RCT identifier and instrument name. Each standardized data file has also been provided in both a comma-separated values file and a *Stata* (version 14; Stata Corp, College Station, TX, USA) *.dta format.

To help users identify the instrument of interest more easily, common data domains were identified to bundle data files according to relevant concepts. Furthermore, variables within the same instrument were made consistent with regard to the variable names and corresponding labels if two or more RCTs used the same instrument. The structures of the standardized data were determined based on the feedback from the

advisory board members consisting of clinical trialists from some of the original NIDA-sponsored RCTs and other substance use disorder research experts. The quality of the standardized data is monitored through routine checks and as-needed basis.

Codebooks and supplemental documentations

For some RCTs, the NIDA Data Share website provided information along with codebooks that describe the definitions of each variable (e.g. CTN0037, CTN0044), whereas for other RCTs, only annotated questionnaires were provided as a guide to using the data (e.g. CTN0001, CTN0004). These supplemental materials were used for identifying variables, developing uniform variable names, and consistent variable and value labels. The standardized codebooks were also produced with the *codebook* command in *Stata*, version 15 (Stata Corp, College Station, TX, USA). Furthermore, we have developed several supplemental documents, including (1) a user's guide which documents processing notes for standardized RCT data, (2) an instrument concordance table, which allows users to identify which instruments are comparable across different RCTs, (3) a variable concordance table, which allows users to identify which variables are comparable across different RCTs, and (4) an assessment schedule table which tabulates assessment schedule for main outcome instruments (e.g. Addiction Severity Index, urine toxicology results) in each study.

Results

Since the inception of this pre-statistical data harmonization initiative in September 2018, we have developed a beta version of the standardized database of 36 NIDA-funded substance use disorder RCTs and produced standardized codebooks and supplemental documentation, including a user's guide, concordance tables (for instruments and variables), and assessment schedule table.

Standardized data structure

Figure 1 presents the data structure of the standardized data. An individual RCT comes with a number of instruments, each of which has been stored in a separate data file. These instruments include not only standardized instruments, such as the Addiction Severity Index and Clinical Opiate Withdrawal Scale, but also non-standardized data elements, such as demographics forms that vary in content across studies. The number of instruments ranged from 16 to 48 per RCT. Each instrument is provided in a separate data file, and data files are named in a consistent manner. For example, the data file for the Addiction Severity Index in CTN0001 is named "CTN0001_ASI." The list of

Table 1. List of the studies included in the data harmonization initiative.

NIDA study code	Study title	N
CSP999	A multicenter clinical trial of buprenorphine in treatment of opiate dependence	736
CSP1008A	A multicenter efficacy/safety trial of buprenorphine/naloxone for the treatment of opiate dependence	326
CSP1019	Double-blind, placebo-controlled trial of selegiline transdermal system (STS) for the treatment of cocaine dependence	300
CSP1020	A phase 3 placebo-controlled, double-blind multi-site trial of lofexidine for opiate withdrawal	68
CSP1021	Double-blind, placebo-controlled multicenter trial of baclofen for the treatment of cocaine dependence	160
CSP1022	Phase 2, double-blind, placebo-controlled trial of STS as an aid for smoking cessation	246
CSP1025	Phase 2, double-blind, placebo-controlled trial of topiramate for the treatment of methamphetamine dependence	140
CSP1026	Phase 2, double-blind, placebo-controlled trial of modafinil for methamphetamine dependence	210
CTN0001	Buprenorphine/naloxone versus clonidine for inpatient opiate detoxification	113
CTN0002	Buprenorphine/naloxone versus clonidine for outpatient opiate detoxification	230
CTN0003	Suboxone (buprenorphine/naloxone) taper: a comparison of two schedules	516
CTN0004	Motivational enhancement therapy (MET) to improve treatment engagement and outcome in subjects seeking treatment for substance abuse	461
CTN0005	Motivational interviewing (MI) to improve treatment engagement and outcome in subjects seeking treatment for substance abuse	423
CTN0006	Motivational incentives for enhanced drug abuse recovery: drug free clinics	454
CTN0007	Motivational incentives for enhanced drug abuse recovery: methadone clinics	403
CTN0009	Smoking cessation treatment with transdermal nicotine replacement therapy in substance abuse rehabilitation program	225
CTN0010	Buprenorphine/naloxone-facilitated rehabilitation for opioid-dependent adolescents/young adults	154
CTN0013	MET to improve treatment utilization and outcome in pregnant substance users	200
CTN0014	Brief strategic family therapy (BSFT) for adolescent drug abusers	482
CTN0015	Women's treatment for trauma and substance use disorders	353
CTN0021	MET to improve treatment engagement and outcome for Spanish-speaking individuals seeking treatment for substance abuse	436
CTN0029	A pilot study of osmotic-release methylphenidate in initiating and maintaining abstinence smokers with ADHD	225
CTN0030	A two-phase randomized controlled clinical trial of buprenorphine/naloxone treatment plus individual drug counseling for opioid analgesic dependence	665
CTN0030A3	Long-term follow-up to CTN0030 (a two-phase RCTs of buprenorphine/naloxone treatment plus individual drug counseling for opioid analgesic dependence)	375
CTN0031	Stimulant abuser groups to engage in 12-step: evaluation of a combined individual-group intervention to reduce stimulant and other drug use by increasing 12-step involvement	471
CTN0037	Stimulant reduction intervention using dosed exercise (stride)	302
CTN0044	Web-delivery of evidence-based, psychosocial treatment for substance use disorders	507
CTN0046	Smoking cessation and stimulant treatment (s-cast): evaluation of the impact of concurrent outpatient smoking cessation and stimulant treatment on stimulant-dependence outcomes	538
CTN0047	Screening, assessment, referral, and treatment in emergency departments (SMART-ED)	1285
CTN0048	Cocaine use reduction with buprenorphine (curb)	302
CTN0051	Extended-release naltrexone versus Buprenorphine for opioid treatment	570
CTN0052	A randomized controlled evaluation of buspirone for relapse prevention in adults with cocaine dependence (BRAC)	62
CTO0001	Phase 2, double-blind, placebo-controlled trial of reserpine for the treatment of cocaine dependence	119
CTO0005	Double-blind, placebo-controlled trial of ondansetron for the treatment of cocaine dependence	65
CTO0007	Phase 2, double-blind, placebo-controlled trial of cabergoline for the treatment of cocaine dependence	140
MDS0004	Phase 2, double-blind, placebo-controlled trial of modafinil for the treatment of cocaine dependence	210
MDS0007	Phase 2, double-blind, placebo-controlled trial of bupropion for methamphetamine dependence	205

NIDA: National Institute on Drug Abuse.

abbreviated names for each instrument (e.g. ASI for Addiction Severity Index) is provided in one of the supplemental documents, the instrument concordance table, which is described below.

All the data files for each RCT have been consistently categorized into eight data domains: (1) *enrollment*, containing information regarding eligibility criteria for RCT participation; (2) *demographics*,

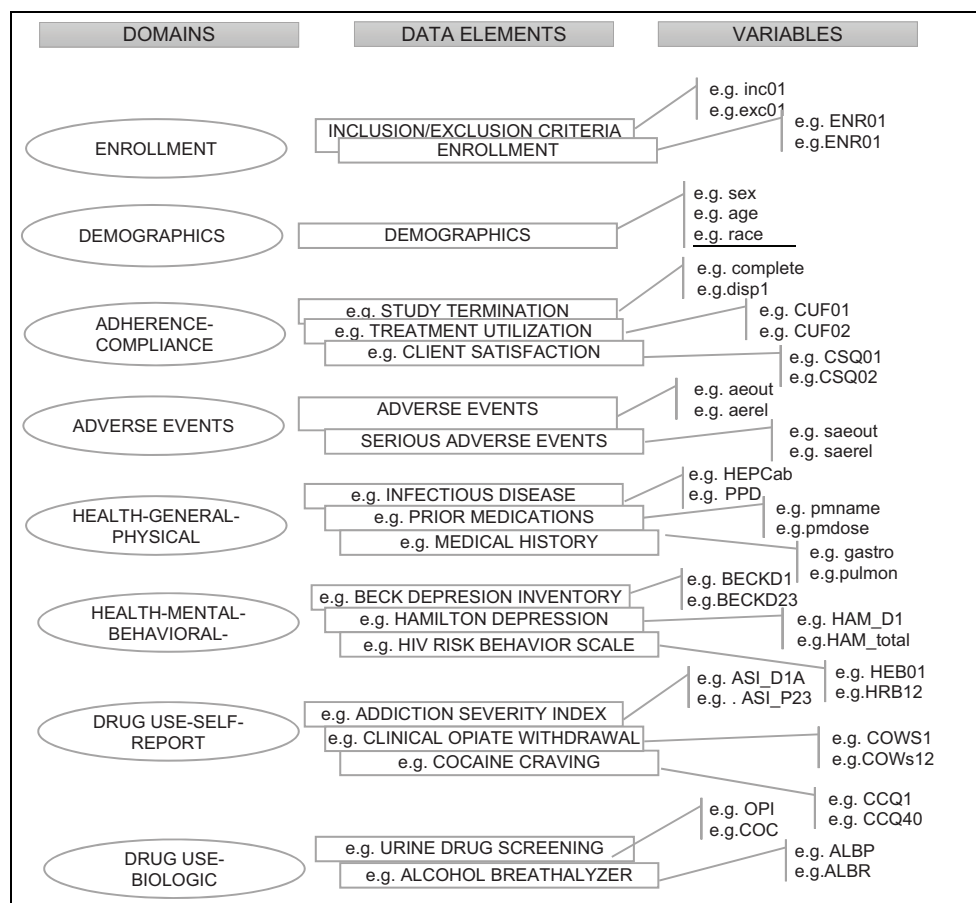


Figure 1. Structure of harmonized data.

containing basic demographic information, such as age, sex, and race/ethnicity; (3) *adherence*, containing information regarding the number of visits made and self-reported measures of adherence with treatments; (4) *adverse events*, containing information about medication- and therapy-related adverse events and often divided into serious and non-serious adverse events; (5) *physical health measures*, containing basic physical health information, such as height, weight, and blood pressure as well as information on medical history and prior/concomitant medications; (6) *mental-behavioral-cognitive health measures*, containing information regarding psychological well-being based on standardized tools, such as Beck Depression Inventory and 36-Item Short Form Survey; (7) *self-reported substance use measures*, containing information based on self-administered tools, such as Addiction Severity Index and Time Line Follow Back Scale; and (8) *biologic substance use measures*, containing information based on objective measures of substance use, such as urine toxicology and alcohol breathalyzer. Data for primary outcome measures in each RCT (e.g. target drug use in the past 30 days, urine toxicology test results) are mostly included in *self-reported substance use* or *biologic substance use measures*. All instruments assessed in each

RCT have been categorized into one of the eight data domains defined above.

Data presentation of variables in each data file has been made consistent across different RCTs with a build-in label and brief definition for each variable. The data files were created so that each data file starts with seven variables that are consistent across all data files and unique for each participant (Figure 2). The first variable, “studyid,” represents the NIDA study identification code (e.g. CTN0001). The second variable, “usubjid,” represents the de-identified subject identification code. All data files from an RCT can be merged by simply using these two variables. The third variable, “arm,” represents the group to which each patient was assigned. Names for treatment arms were taken directly from original data files. We created variables “arm1” (treatment arm for phase 1) and “arm2” (treatment arm for phase 2) for study NIDA-CTN0030/0030A, which was a two-step trial. For recruited individuals who were not randomized but had baseline assessments, “arm” was coded either as “not randomized” or left missing, depending on availability of information in original data files. The fourth variable, “assessdays,” represents days since randomization (e.g. −3 = 3 days before randomization) for each assessment. This

CTO0005 UDS data file

Study ID
"studyid"

Subject ID
"usubjid"

Treatment arm
"arm"

Days since randomization
"assessdays"

Study specific visit ID
"visno"

Study phase
"phase"

Measures/instruments
"measure"

	studyid	usubjid	arm	assessdays	visno	phase	measure	D1A	D1B	D1C	D2A	D2B	D2C	D3A
1	CTO0005	05_4891	4.0 mg Ondansetron	-3	SCRNBASE	BASELINE	ADDICTION SEVERITY INDEX LITE	28	20	.	0	5	.	0
2	CTO0005	05_4891	4.0 mg Ondansetron	28	WEEK4V1	ACTIVE	ADDICTION SEVERITY INDEX LITE	18	20	.	0	1	.	0
3	CTO0005	05_4891	4.0 mg Ondansetron	60	WEEKSV3	ACTIVE	ADDICTION SEVERITY INDEX LITE	12	20	.	0	1	.	0
4	CTO0005	05_8342	4.0 mg Ondansetron	-2	SCRNBASE	BASELINE	ADDICTION SEVERITY INDEX LITE	20	25	.	0	0	.	0
5	CTO0005	05_8342	4.0 mg Ondansetron	61	WEEKSV3	ACTIVE	ADDICTION SEVERITY INDEX LITE	13	23	.	0	0	.	0
6	CTO0005	05_12200	Placebo	-14	SCRNBASE	BASELINE	ADDICTION SEVERITY INDEX LITE	10	16	.	10	5	.	0
7	CTO0005	05_12200	Placebo	31	WEEK4V1	ACTIVE	ADDICTION SEVERITY INDEX LITE	6	15	.	3	0	.	0
8	CTO0005	05_36247	Placebo	-10	SCRNBASE	BASELINE	ADDICTION SEVERITY INDEX LITE	8	8	.	7	6	.	0
9	CTO0005	05_36247	Placebo	37	WEEK4V1	ACTIVE	ADDICTION SEVERITY INDEX LITE	10	6	.	6	4	.	0
10	CTO0005	05_56602	0.25 mg Ondansetron	-12	SCRNBASE	BASELINE	ADDICTION SEVERITY INDEX LITE	2	10	.	0	10	.	0
11	CTO0005	05_100607	Placebo	-8	SCRNBASE	BASELINE	ADDICTION SEVERITY INDEX LITE	14	23	.	7	1	.	0
12	CTO0005	05_100607	Placebo	26	WEEK4V1	ACTIVE	ADDICTION SEVERITY INDEX LITE	5	23	.	0	6	.	0
13	CTO0005	05_100607	Placebo	62	WEEKSV3	ACTIVE	ADDICTION SEVERITY INDEX LITE	2	23	.	0	6	.	0
14	CTO0005	05_105642	Not randomized	.	SCRNBASE	BASELINE	ADDICTION SEVERITY INDEX LITE	4	6	.	0	6	.	0
15	CTO0005	05_107072	0.25 mg Ondansetron	-13	SCRNBASE	BASELINE	ADDICTION SEVERITY INDEX LITE	10	30	.	0	20	.	0
16	CTO0005	05_107072	0.25 mg Ondansetron	30	WEEK4V1	ACTIVE	ADDICTION SEVERITY INDEX LITE	10	30	.	6	28	.	0
17	CTO0005	05_130936	1.0 mg Ondansetron	-8	SCRNBASE	BASELINE	ADDICTION SEVERITY INDEX LITE	12	23	.	1	10	.	0
18	CTO0005	05_130936	1.0 mg Ondansetron	25	WEEK4V1	ACTIVE	ADDICTION SEVERITY INDEX LITE	20	25	.	1	10	.	0
19	CTO0005	05_130936	1.0 mg Ondansetron	46	WEEKSV3	ACTIVE	ADDICTION SEVERITY INDEX LITE	15	25	.	1	10	.	0
20	CTO0005	05_133137	0.25 mg Ondansetron	-11	SCRNBASE	BASELINE	ADDICTION SEVERITY INDEX LITE	0	16	.	0	4	.	1
21	CTO0005	05_133137	0.25 mg Ondansetron	28	WEEK4V1	ACTIVE	ADDICTION SEVERITY INDEX LITE	1	15	.	1	15	.	0
22	CTO0005	05_135020	0.25 mg Ondansetron	-13	SCRNBASE	BASELINE	ADDICTION SEVERITY INDEX LITE	20	26	.	20	26	.	0
23	CTO0005	05_146359	Not randomized	.	SCRNBASE	BASELINE	ADDICTION SEVERITY INDEX LITE	3	30	.	0	0	.	0
24	CTO0005	05_151710	0.25 mg Ondansetron	-8	SCRNBASE	BASELINE	ADDICTION SEVERITY INDEX LITE	0	0	.	0	0	.	0
25	CTO0005	05_151710	0.25 mg Ondansetron	28	WEEK4V1	ACTIVE	ADDICTION SEVERITY INDEX LITE	0	0	.	0	0	.	0
26	CTO0005	05_171483	Not randomized	.	SCRNBASE	BASELINE	ADDICTION SEVERITY INDEX LITE	20	28	.	15	12	.	0
27	CTO0005	05_176622	1.0 mg Ondansetron	-8	SCRNBASE	BASELINE	ADDICTION SEVERITY INDEX LITE	14	9	.	14	9	.	0
28	CTO0005	05_176622	1.0 mg Ondansetron	50	WEEKSV3	ACTIVE	ADDICTION SEVERITY INDEX LITE	12	9	.	9	9	.	0
29	CTO0005	05_182325	Placebo	-4	SCRNBASE	BASELINE	ADDICTION SEVERITY INDEX LITE	6	20	.	6	20	.	0
30	CTO0005	05_189528	0.25 mg Ondansetron	-7	SCRNBASE	BASELINE	ADDICTION SEVERITY INDEX LITE	6	2	.	2	2	.	0
31	CTO0005	05_192852	4.0 mg Ondansetron	-9	SCRNBASE	BASELINE	ADDICTION SEVERITY INDEX LITE	8	12	.	0	0	.	0
32	CTO0005	05_192852	4.0 mg Ondansetron	31	WEEK4V1	ACTIVE	ADDICTION SEVERITY INDEX LITE	8	15	.	0	0	.	0
33	CTO0005	05_195100	0.25 mg Ondansetron	-12	SCRNBASE	BASELINE	ADDICTION SEVERITY INDEX LITE	0	15	.	0	0	.	0
34	CTO0005	05_195100	0.25 mg Ondansetron	38	WEEKSV3	ACTIVE	ADDICTION SEVERITY INDEX LITE	2	0	.	0	0	.	0
35	CTO0005	05_301031	4.0 mg Ondansetron	-14	SCRNBASE	BASELINE	ADDICTION SEVERITY INDEX LITE	15	10	.	0	0	.	0

Figure 2. Harmonized data presentation of variables in each data file (e.g. CTO0005, Addiction Severity Index-Lite data file).

variable helps users identify which measures were assessed when (e.g. cocaine urine toxicology tests were performed 3 days before randomization and 14 days and 52 days after randomization). The fifth variable, "visno" represents RCT-specific assessment visit number (e.g. WEEK4 V1 indicates visit number 1 in Week 4). The sixth variable, "phase," represents the phase of the study, which was derived based on "visno" to allow users to distinguish whether assessments were recorded before the active treatment ("BASELINE"), during active treatment ("ACTIVE"), or after active treatment ("FOLLOWUP"). Exceptionally for CTN0030/0030A, which was a two-step trial, we created two values for active phase, "ACTIVE1" (during phase 1 treatment) and "ACTIVE2" (during phase 2 treatment). The seventh variable, "measure," represents the name of the instruments that are stored in a data file (e.g. ADDICTION SEVERITY INDEX-LITE). Data files for most of the measures have been structured in a long (longitudinal) format in which multiple assessments are recorded vertically across different phases of the study. Because in many cases variables "assessdays," "visno," and "phase" were not all available for every assessment due to missing information in original data files, a variable "sequence" (integer starting from 1) was created

for some instruments, using available information that allows users to identify the sequence of assessments. These standardized variables allow users to combine multiple datasets from the RCTs and also to conduct basic analyses using each data file separately.

Codebooks and supplemental documentations

Standardized codebooks were produced using information from various documents for each RCT, including annotated questionnaires, protocols, and spreadsheets of data dictionaries. For all variables, the codebooks consist of variables names with brief descriptions, including mode of data collection (e.g. laboratory test, self-report), variable type (string/numeric), and the number of missing values. For string variables (e.g. patient id), the number of unique values is also presented. For numeric continuous variables (e.g. quantitative measures from a urine sample), mean, standard deviation, and percentiles are presented. For numeric categorical variables (e.g. qualitative measures from a urine sample), a range of values and variable labels for associated numeric values are presented.

Several supplemental documents were also created. These included a *user's guide*, which briefly describes

domain	measure	shortname	NIDA-CSF-999	NIDA-CSF-1008	NIDA-CSF-1019	NIDA-CSF-1020	NIDA-CSF-1021	NIDA-CSF-1022	NIDA-CSF-1025	NIDA-CSF-1026	NIDA-CSF-1001	NIDA-CSF-1002	NIDA-CSF-1003	NIDA-CSF-1004	NIDA-CSF-1005	NIDA-CSF-1006	NIDA-CSF-1007	NIDA-CSF-1009	NIDA-CSF-1010	NIDA-CSF-1013	NIDA-CSF-1014	NIDA-CSF-1015	NIDA-CSF-1021	NIDA-CSF-1029	NIDA-CSF-1031	NIDA-CSF-1037	NIDA-CSF-1046	NIDA-CSF-1051	NIDA-CSF-1052	NIDA-CSF-1001	NIDA-CSF-1007	NIDA-MOS-000
ADHERENCE/COMPLIANCE	CLIENT SATISFACTION QUESTIONNAIRE	CSQ																														
ADHERENCE/COMPLIANCE	COMPLIANCE	COMP																														
ADHERENCE/COMPLIANCE	DOSE ADMINISTRATION RECORD	DOSEA	X			X	X																									
ADHERENCE/COMPLIANCE	FOLLOW-UP ASSESSMENT	FOLLOWUP	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
ADVERSE EVENTS	ADVERSE EVENTS	ADVERSE	X	X																												
ADVERSE EVENTS	ADVERSE EVENTS AND CONCOMITANT MEDICATIONS	AECON		X	X																											
ADVERSE EVENTS	SERIOUS ADVERSE EVENTS	SAE	X	X	X	X	X	X																								
DEMOGRAPHICS	DEMOGRAPHICS	DEM	X	X	X	X	X	X																								
DRUG USE - BIOLOGIC	ALCOHOL BREATHALYZER	ALB																														
DRUG USE - BIOLOGIC	URINE DRUG SCREEN - AMPHETAMINES	UDS	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
DRUG USE - BIOLOGIC	URINE DRUG SCREEN - COCAINE	UDS	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
DRUG USE - BIOLOGIC	URINE DRUG SCREEN - OPIATES	UDS		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
DRUG USE - BIOLOGIC	URINE DRUG SCREEN - OXYCODONE	UDS											X																			
DRUG USE - SELF REPORT/SUE	ADDICTION SEVERITY INDEX LITE	ASI			X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
DRUG USE - SELF REPORT/SUE	ADJECTIVE RATING SCALE FOR WITHDRAWAL	ARSW										X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
DRUG USE - SELF REPORT/SUE	ASI LITE DRUG/ALCOHOL USE	SU																														
DRUG USE - SELF REPORT/SUE	SUBSTANCE USE REPORT	SUR	X	X	X	X	X	X	X																							
DRUG USE - SELF REPORT/SUE	SUBSTANCE USE-CALENDAR ASSESSMENT	CALENDAR																														
ENROLLMENT/RANDOMIZATION	INCLUSION/EXCLUSION CRITERIA	INCEXC			X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
HEALTH - GENERAL/PHYSICAL	CHEMISTRIES	CHEM	X	X	X	X	X	X	X									X														
HEALTH - GENERAL/PHYSICAL	CONCOMITANT MEDICATIONS	CONMEDS	X									X																				
HEALTH - GENERAL/PHYSICAL	ELECTROCARDIOGRAM	ECG	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
HEALTH - GENERAL/PHYSICAL	HEMATOLOGY	HEMAT	X	X	X	X	X	X	X																							
HEALTH - GENERAL/PHYSICAL	INFECTIOUS DISEASE ASSESSMENT	INFECDIS	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
HEALTH - GENERAL/PHYSICAL	MEDICAL HISTORY	MEDHIST	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
HEALTH - MENTAL/BEHAVIOR	BECK DEPRESSION INVENTORY	BECK																X														
HEALTH - MENTAL/BEHAVIOR	HAMILTON DEPRESSION RATING SCALE	HAMD		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
HEALTH - MENTAL/BEHAVIOR	HIV RISK BEHAVIOR SCALE	HRBS																														
HEALTH - MENTAL/BEHAVIOR	SCID WORKSHEET	SCID		X	X	X	X	X	X	X				X	X	X	X						X		X		X		X	X	X	

Figure 3. Excerpt from data concordance table.

how the harmonized data files were constructed, and how instruments and variables of interest can be found for secondary data analysis. The *user's guide* also describes how data files can be merged with each other by providing several examples. Second, we have developed a *data concordance table* that helps users to identify which instruments are categorized in each of the eight data domains defined above. This document is especially useful since the number of instruments collected in each RCT varied (Figure 3). The *data concordance table* allows users to easily identify comparable instruments across different RCTs. Third, we have created a *variable concordance table* since some items were omitted from some of the instruments in some RCTs (e.g. quantitative measures for urine toxicology tests were only available for a subset of RCTs). The *variable concordance table* presents the item-level concordance within each instrument. Fourth, we have created an *assessment schedule table* showing the timing and frequencies of assessment for major substance use disorder outcome measures, such as urine toxicology tests and the Addiction Severity Index for each RCT (Figure 4).

Examples of secondary data analyses

This section aims to illustrate the utility of the harmonized datasets by providing several examples of secondary data analyses. Table 2 presents basic descriptive statistics of 36 RCTs. Standardized data allow researchers to easily pool data from multiple RCTs and conduct comparative studies across RCTs. Secondary analyses leveraging the harmonized data have been published by our investigative team, a group of researchers who were not involved in primary data collection of the NIDA-funded RCTs.^{18–24} First, a series of studies by Susukida

and colleagues^{18–22} examined the generalizability of the findings from substance use disorder RCTs to the target populations. These studies used harmonized data from a subset of the NIDA-funded RCTs described above and compared the characteristics of individuals participating in substance use disorder RCTs with individuals receiving treatment in usual care settings. The main findings of these studies were that individuals recruited into substance use disorder RCTs appear to differ from individuals receiving treatment in usual care settings. Specifically, RCT participants had more years of education and a greater likelihood of full-time work compared with people receiving care in usual care settings. In a further step, statistical weighting was used to recompute the effects from these substance use disorder RCTs such that the RCT participants had characteristics that resembled those of patients in the target populations.²⁵ Such re-weighting of the samples changed the treatment effects in a number of the RCTs. Most commonly, the positive effects of trials became statistically non-significant after re-weighting the RCT sample to match the target population.

Second, we conducted a study assessing the validity of the psychiatric problems subscale of the Addiction Severity Index,²³ a detailed measure commonly used in substance use disorder treatment research to detect psychiatric comorbidity using pooled harmonized data from 11 NIDA-funded RCTs involving 1660 participants. Our results demonstrated moderate accuracy of this Addiction Severity Index subscale in detecting the presence of any serious mental disorder against the gold standard of diagnoses of serious mental illnesses based on structured or semi-structured interviews. The results support the utility of this Addiction Severity Index subscale in screening psychiatric comorbidity among patients receiving substance use disorder treatments.

Cont. Table 2: Study schema of RCTs on stimulant use disorder and the assessment points by instruments																																				
Study ID	Instrument	Study week																																		
		-	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	28	36 (9m)	48	52 (12m)					
MDS4	Phase	B	A												F																					
	ASI		√													√																				
	UDS*		3√	3√	3√	3√	3√	3√	3√	3√	3√	3√	3√	3√	3√				√																	
	RBM		√													√																				
CSP1021	Phase	B	A								F																									
	ASI		√				√				√																									
	UDS*		3√	3√	3√	3√	3√	3√	3√	3√	3√		√		√																					
	RBM		√									√																								
CSP1025	Phase	B	A												F																					
	ASI		√				√				√			√				√																		
	UDS		3√	3√	3√	3√	3√	3√	3√	3√	3√	3√	3√	3√				√																		
	RBM		√													√																				
CSP1026	Phase	B	A												F																					
	ASI		√													√																				
	UDS**		3√	3√	3√	3√	3√	3√	3√	3√	3√	3√	3√	3√				√																		
	RBM		√													√																				
CTN31	Phase	B	A								F																									
	ASI		√												√																					
	UDS		√	√				√					√																							
	RBM																									√										

Figure 4. Excerpt from assessment schedule table.

B: Baseline/Screening phase; A: Active treatment phase; F: Follow-up phase; ASI: Addiction Severity Index; UDS: Urine Drug Screening; RBM: Risk Behavior Measures.

*Three times a week for benzoylecgonine (BE) and creatinine and once a week for the other drugs.

**Three times a week for methamphetamine and creatinine. Other drugs were examined only at baseline.

Furthermore, in collaboration with a biostatistician, we have conducted a study to examine precision and power gains associated with adjustment for baseline variables in stratified RCTs, using the harmonized data of three stratified NIDA RCTs.²⁴ The results demonstrated that the variance of the treatment estimates reduced up to 35% by adjusting for baseline variables, indicating that researchers planning to perform adjustment for strata and additional baseline variables could achieve the same precision with approximately 35% fewer participants.

Discussion

Pre-statistical data harmonization was used to standardize de-identified primary data from substance use disorder treatment RCTs funded by the NIDA. Our data harmonization initiative is in line with a growing movement to promote data dissemination and transparency practices in scientific research communities.^{1,2} Consistent with this movement, the NIDA Clinical Trials Network has initiated efforts to make data from completed RCTs of various substance use disorder treatments publicly available through the NIDA Data Share website; however, the lack of standardized data products is a major barrier to further use of these valuable data, which are currently underutilized. We have created a harmonized database of variables across 36

RCTs with a build-in label and a brief definition for each variable. Data files from the RCTs have been consistently categorized into eight clinically relevant data domains. Furthermore, we have developed standardized codebooks and concordance tables to help identify instruments and variables of interest in each RCT more easily.

The scope of this data harmonization initiative was limited to the RCT data that were available from the NIDA Data Share website as of September 2018. Since the inception of the project, data for two additional RCTs (i.e. CTO0011 and CTO0012) were deposited on the website. The harmonization of these additional RCT data and future RCT data would require additional support which we anticipate will come from private or public funding agencies. We expect that the beta version of the standardized database of 36 NIDA-funded substance use disorder RCTs will serve as a proof of concept to make the case for future support, by highlighting the importance of increasing usability of individual patient data from previously conducted substance use disorder RCTs.

Our data harmonization initiative focused on flexible “pre-statistical” data harmonization involving standardization of data collected through diverse data collection tools.^{16,17} While this data standardization approach has strengths such that it can be flexibly

Table 2. Characteristics of the trials and participants.

Study ID	Arms	Treatment				Sample size (n)	Female (%)	Age (mean)	White (%)	Married (%)
		Treatment 1	Treatment 2	Treatment 3	Treatment 4/ placebo					
CSP999	Bup (1 mg)	Bup (4 mg)	Bup (8 mg)	Bup (16 mg)	736	32.5	36.0	48.8	26.4	
CSP1008A	Bup-Nx	BU	-	Placebo	326	35.0	37.6	60.4	22.2	
CSP1019	Selegiline	-	-	Placebo	300	22.0	40.7	26.5	18.7	
CSP1020	Lofexidine	-	-	Placebo	68	13.2	41.3	51.5	17.6	
CSP1021	Baclofen	-	-	Placebo	160	20.6	42.7	25.6	20.6	
CSP1022	Selegiline	-	-	Placebo	246	49.2	46.5	68.7	43.5	
CSP1025	Topiramate	-	-	Placebo	140	36.4	38.0	77.9	18.6	
CSP1026	Modafinil 200	Modafinil 400	-	Placebo	210	40.9	39.0	73.3	21.9	
CTN0001	Bup-Nx	Clonidine	-	-	113	39.8	36.2	55.7	23.0	
CTN0002	Bup-Nx	Clonidine	-	-	230	28.3	38.8	40.0	15.7	
CTN0003	Bup-Nx (7-d taper)	Bup-Nx (28-d taper)	-	-	516	33.0	35.9	76.5	24.2	
CTN0004	MET	-	-	ST	461	29.1	35.3	42.0	15.6	
CTN0005	MI	-	-	ST	423	42.1	33.3	71.9	17.0	
CTN0006	Incentives	-	-	TAU	454	54.9	35.9	35.7	18.5	
CTN0007	Incentives	-	-	TAU	403	43.9	41.9	25.6	10.9	
CTN0009	SmC	-	-	TAU	225	48.0	42.1	36.0	18.7	
CTN0010	Bup-Nx (12 week)	Bup-Nx (14 days)	-	-	154	41.6	19.7	70.1	5.2	
CTN0013	MET	-	-	ST	200	100.0	26.2	37.2	14.5	
CTN0014	BFST	-	-	TAU	482	21.4	16.0	40.8	-	
CTN0015	SS	WHE	-	-	353	100.0	39.2	48.9	15.9	
CTN0021	MET	-	-	CAU	436	10.8	32.7	0.0	29.4	
CTN0029	ORMeth	-	-	Placebo	255	43.5	37.8	80.4	-	
CTN0030	Bup-Nx + SC	Bup-Nx + EIC	-	-	665	39.6	33.2	90.8	28.6	
CTN0031	STAGE-12	-	-	TAU	471	58.8	38.3	50.3	12.5	
CTN0037	Exercise	Health education	-	-	302	40.1	38.5	45.4	13.2	
CTN0044	TES	-	-	TAU	507	37.9	35.4	57.6	14.2	
CTN0046	TAU + SmC	-	-	TAU	538	48.0	37.0	59.5	10.8	
CTN0047	BS + B	SAR	MS	-	1285	30.1	36.4	38.0	9.5	
CTN0051	Nalt-ER	Bup-Nx	-	-	570	33.9	33.9	69.1	9.1	
CTN0052	Bupirone	-	-	Placebo	62	37.1	46.1	25.8	12.9	
CTO0001	Reserpine	-	-	Placebo	119	29.4	41.0	21.0	17.8	
CTO0005	Ondansetron (0.25 mg)	Ondansetron (1 mg)	Ondansetron (4 mg)	-	65	15.4	36.2	38.5	20.0	
CTO0007	Cabergoline	-	-	Placebo	140	15.0	40.1	37.9	25.0	
MDS0004	Modafinil 200	Modafinil 400	-	Placebo	210	28.1	41.8	39.2	18.6	
MDS0007	Bupropion	-	-	Placebo	205	35.1	39.3	68.3	15.6	

Bup: Buprenorphine; Bup-Nx: Buprenorphine–Naloxone; BS + B: Brief Intervention plus Booster; BSFT: Brief Strategic Family Therapy; CAU: Counseling as usual; EIC: Enhanced Individual Counseling; Hisp: Hispanic; HS: High School; MET: Motivational Enhancement Therapy; MI: Motivational Interviewing; MS: Minimal Screening; NA: Not-applicable; Nalt-ER: Naltrexone-ER; NH black: Non-Hispanic black; NH white: Non-Hispanic white; ORMeth: Osmotic-Release Methylphenidate; SAR: Screening, Assessment, and Referral; SC: Standard Counseling; SmC: Smoking cessation; ST: Standard treatment; SS: Seeking Safety; TAU: Treatment as usual; TES: Therapeutic Education System; WHE: Women's Health Education.

extended across a broader range of RCTs with different data collection tools and procedures,¹⁶ ultimate data synthesis would require further analytical decision models specified by users to reach inferential equivalence of measures across RCTs. Our data standardization initiative also did not involve collecting analytic files that were used by original investigators to produce results in the primary publications of the RCTs. Previous studies demonstrate that it may be impossible to exactly replicate results in published reports, even when data are available, and the standardized data and codebooks that we have developed might not allow users to replicate published study results.⁷ Over time, with greater exposure and knowledge about the harmonized data and this project, we expect that original investigators will be more willing to contribute their opinions or analytic files to the project. Finally, we expect that it would take time for the data standardization guidelines to be adopted by future trialists conducting similar substance use disorder treatment RCTs. Until the data standardization becomes a routine practice by future trialists, we expect that retrospective data standardization of completed RCTs similar to our initiative would be a mainstream practice. To facilitate adherence to the guidelines by future trialists, we will continue to seek active engagement of the advisory board members of our data harmonization initiative.

A number of limitations of this harmonization initiative should also be considered. First, the project did not involve creating or using an interactive data harmonization platform.¹⁷ However, the harmonized data produced can be translated into such a platform for future meta-analyses. Second, we did not attempt to statistically harmonize measures of the same domain (e.g. different depression instruments).²⁶ However, availability of uniform variable names and value labels for the same instruments across RCTs and categorizing instruments from the same domain would facilitate future statistical harmonization. Third, our data harmonization project was a retrospective initiative. Ideally, data harmonization efforts should be prospective.^{27,28} The development and use of core outcome sets can increase comparability across trials and minimize waste.²⁹ Moreover, prospective harmonization allows for close involvement of primary investigators and integration of the harmonization with primary data management and analyses. As such, core outcome sets and prospective harmonization should be an integral part of future RCTs.

Conclusion

As the country struggles to tackle the growing problem of substance use disorders, and particularly the recent opioid epidemic, there is a pressing need for reproducible and generalizable studies to examine various interventions. The NIDA Data Share website includes data

from numerous studies involving thousands of patients treated for various substance use disorders. Secondary analyses of these data can provide useful information by addressing new questions about treatment of substance use disorders and formulating new hypotheses. Our pre-statistical data harmonization initiative has made these data more accessible and user-friendly to a wide circle of researchers, which would encourage various secondary data analyses and help advance the standard of research and ultimately, potentially treatment of various substance use disorders. The societal benefits from the deliverables from this data harmonization initiative are expected to be quite large and timely in the face of critical public health concerns in our nation. Future directions and next steps include the effective dissemination of the harmonized data to encourage secondary data analyses in the broader scientific community.

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